

EAI 6010 Module 6 Project Guide

Iowa Liquor Sales 2025 | GPT-4o Evaluation

1. Dataset

Metric	Value
Name	Iowa Liquor Sales 2025
Source	data.iowa.gov/catalog/dataset/1262
Size	2,490,620 rows 21 columns Jan - Dec 2025
Total Revenue	\$424,834,283.15
Bottles Sold	30,441,277
Stores	2,300 Products: 4,356 Counties: 103 Vendors: 234

2. How to Run the Notebook

Step 1 - Download the Dataset

- Go to: data.iowa.gov/catalog/dataset/1262
- Click Export and download as CSV (approx. 538 MB)

Step 2 - Open Google Colab

- Go to colab.research.google.com and create a new notebook

Step 3 - Get an OpenAI API Key

- Go to platform.openai.com/api-keys and click Create new secret key
- Add at least \$5 in billing at platform.openai.com/settings/billing
- 5 questions cost less than \$0.05 total

Note: Never share your API key publicly. Paste it only inside your Colab notebook.

Step 4 - Run Each Section in Order

Section 1: Install libraries

```
!pip install openai prophet plotly -q
```

Section 2: Upload and load data

```
from google.colab import files uploaded = files.upload() # select your CSV df =  
pd.read_csv("iowa_liquor_sales_2025_1262_rows.csv", low_memory=False) df["ordered_on"] =  
pd.to_datetime(df["ordered_on"])
```

Note: A file picker will appear. Upload takes 2-3 minutes due to file size (538 MB).

Section 3: EDA - run dataset overview, top 10 tables, monthly chart. No API key needed.

Section 4: GPT-4o Evaluation - paste your API key, then run the 5-question evaluation.

Section 5: Ground Truth - calculates correct pandas answers and builds evaluation table.
Section 6: Prediction - Linear Regression model forecasts June 2026 revenue.
Section 7: Summary - prints all key findings.

3. Key Findings

EDA

Metric	Value
Top Store	HY-VEE #3 / BDI / DES MOINES (\$13,668,737.03)
Top Product	TITO'S HANDMADE VODKA (\$29,952,776.16)
Top County	POLK (\$97,441,531.84)
Lowest Top-10 County	CLINTON (\$10,718,074.08)
Peak Month	December 2025 (\$41,662,102.00)
Lowest Month	January 2025 (\$31,111,677.66)
Jan to Dec Growth	+33.9%

GPT-4o Evaluation Results

Q#	Question	Ground Truth	Match?
Q1	Top store by revenue?	HY-VEE #3 / BDI / DES MOINES (\$13,668,737.03)	Yes
Q2	Top product by revenue?	TITO'S HANDMADE VODKA (\$29,952,776.16)	Yes
Q3	Monthly trend direction?	Increased +33.9% (Jan \$31.1M to Dec \$41.7M)	Yes
Q4	Lowest top-10 county?	CLINTON (\$10,718,074.08)	Yes
Q5	Anomalies in monthly data?	December 2025 (\$41,662,102)	Partial

GPT-4o Performance

Metric	Value
Accuracy	4/5 questions correct (80%)
Avg Latency	1.47 seconds per question
Total Tokens Used	5,916 (avg 1,183 per question)
Max Raw Rows Processable	~1,600 to 2,560 rows
Dataset Coverage	Less than 0.1% of 2,490,620 rows
Conclusion	Pre-aggregation required. Not sufficient for production use.

June 2026 Revenue Prediction (Linear Regression)

Metric	Value
Predicted June 2026 Revenue	\$42,899,748
Monthly Growth Rate	\$651,904 per month
R-Squared Score	0.5942
Limitation	Linear model cannot capture seasonal spikes

4. Paper Outline (Snigdha)

1,000 words max | APA format | Times New Roman 12pt | Double spaced

Q1 - What ML service and why?

- Service: OpenAI GPT-4o API
- Why: Best-in-class LLM for natural language Q&A on structured data
- Dataset: Iowa Liquor Sales 2025 - 2,490,620 rows, \$424.8M total revenue

Q2 - How well does it perform?

- Accuracy: 80% (4/5 correct on summarized data)
- Latency: 1.47s avg per question | Tokens: 5,916 total
- Critical limit: GPT-4o handles max ~1,600 raw rows vs 2.49M in dataset
- Verdict: NOT sufficient for production use without pre-aggregation

Q3 - Recommendation: Custom model or AutoML?

- Recommendation: Custom RAG (Retrieval-Augmented Generation) pipeline
- Architecture: Aggregate data with pandas, retrieve summary, pass to GPT-4o
- Why not AutoML: RAG preserves GPT-4o strengths while solving the token limit
- Expected improvement: from 80% to 95%+ accuracy

Suggested Structure

- Introduction (100 words)
- ML Service Selection (200 words)
- Methodology (150 words)
- Results (300 words) - include evaluation table and June 2026 prediction
- Recommendation (200 words)
- Conclusion (50 words)

5. PPT Slide Outline (Lavanya)

Slide	Title	Key Content
1	Title	Project name, team, date
2	Problem Statement	What is an AI Data Analyst Copilot? Why GPT-4o?
3	Dataset Overview	2.49M rows, \$424.8M revenue, 2,300 stores, 4,356 products
4	EDA Findings	Top 10 tables + monthly_revenue_2025.png chart
5	GPT-4o Methodology	How evaluation worked, token limit constraint
6	GPT-4o Results	Evaluation table — 4/5 correct, 80% accuracy, 1.47s avg
7	Key Limitation	Token limit: 1,600 max rows vs 2.49M = less than 0.1% coverage
8	June 2026 Prediction	lr_forecast_june2026.png + \$42,899,748 predicted revenue
9	Recommendation	RAG pipeline: Data to Aggregation to GPT-4o to Answer
10	Next Steps	Streamlit app, RAG implementation, validate June actual data